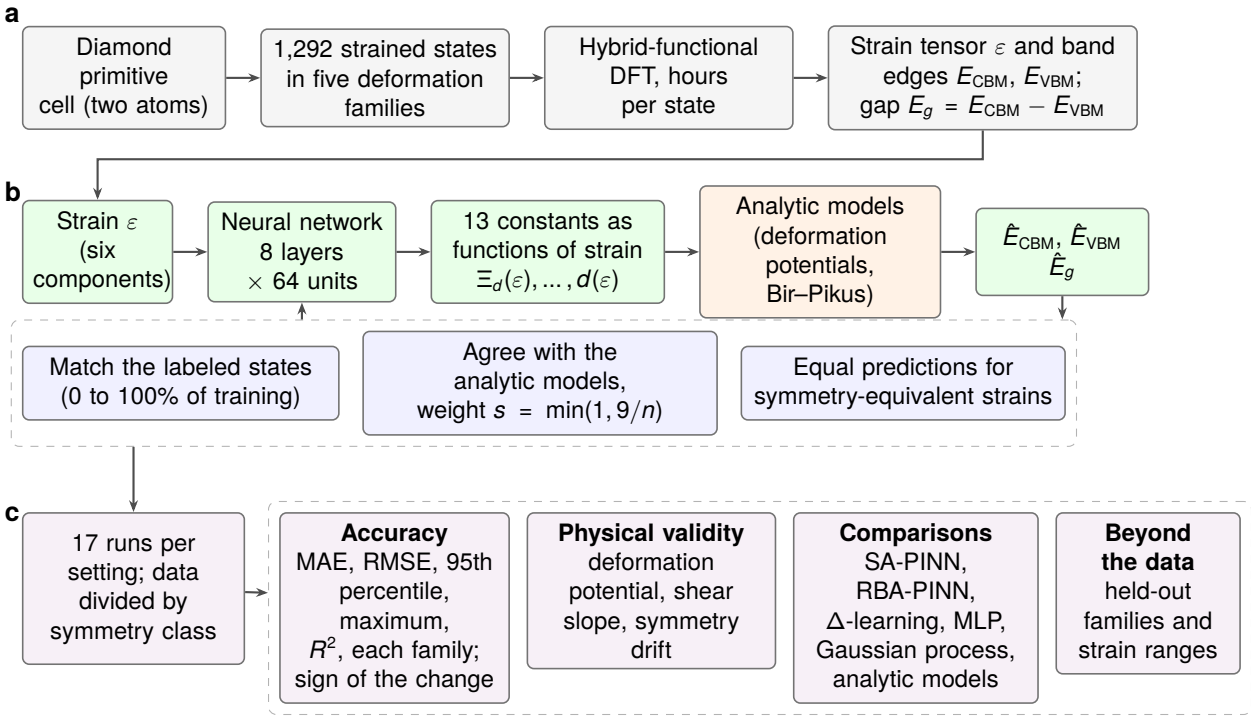

a
Diamond primitive cell (two atoms)
1,292 strained states in five deformation families
Hybrid-functional DFT, hours per state
Strain tensor ε and band edges E_{CBM}, E_{VBM}; gap $E_g = E_{CBM} - E_{VBM}$
b
Strain ε (six components)
Neural network 8 layers × 64 units
13 constants as functions of strain $\Xi_d(\varepsilon), \ldots, d(\varepsilon)$
Analytic models (deformation potentials, Bir–Pikus)
$\hat{E}_{CBM}$, $\hat{E}_{VBM}$ $\hat{E}_g$
Match the labeled states (0 to 100% of training)
Agree with the analytic models, weight $s = \min(1, 9/n)$
Equal predictions for symmetry-equivalent strains
c
17 runs per setting; data divided by symmetry class
Accuracy
MAE, RMSE, 95th percentile, maximum, R^2, each family; sign of the change
Physical validity
deformation potential, shear slope, symmetry drift
Comparisons
SA-PINN, RBA-PINN, Δ-learning, MLP, Gaussian process, analytic models
Beyond the data
held-out families and strain ranges